SWIGGY SALES ANALYSIS

Business Requirements Document

SQL Analytics Project

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1. Data Cleaning & Validation

The raw table **`swiggy_data`** contains food delivery records across states, cities, restaurants, categories, and dishes. The first goal is to ensure data quality through the following validation steps:

1.1 Null Check

Identify and handle missing values in the following critical columns:

Column	Description
State	Geographic state of the order
City	City where the order was placed
Order_Date	Date of the order
Restaurant_Name	Name of the restaurant
Location	Specific location / area
Category	Cuisine or food category
Dish_Name	Name of the ordered dish
Price_INR	Price of the dish in Indian Rupees
Rating	Customer rating of the dish
Rating_Count	Number of ratings received

1.2 Blank / Empty String Check

Detect fields containing blank or empty string values that may cause inaccurate analysis. These should be treated similarly to nulls or standardized as appropriate for downstream processing.

1.3 Duplicate Detection

Find duplicate rows by grouping on all business-critical columns. Identify records that represent the same logical order more than once.

1.4 Duplicate Removal

Use the **`ROW_NUMBER()`** window function to rank duplicate rows and retain only one clean copy for each unique order, deleting surplus duplicates.

2. Dimensional Modelling (Star Schema)

Dimensional modelling organizes data to make analysis simple, consistent, and highly efficient. Instead of keeping all information in one large table, the Star Schema separates descriptive attributes into dimension tables and keeps measurable values in a central fact table. This structure:

- Reduces data duplication and improves clarity
- Enables faster reporting by querying focused, smaller tables
- Aligns with modern analytics and BI tools for smooth dashboard creation
- Provides a clean, scalable, and performance-optimized foundation for analytics

2.1 Dimension Tables

Build the following dimension tables populated with distinct values from the cleaned source:

Dimension Table	Key Attributes
`dim_date`	Year, Month, Quarter, Week
`dim_location`	State, City, Location
`dim_restaurant`	Restaurant_Name
`dim_category`	Cuisine / Category
`dim_dish`	Dish_Name

2.2 Fact Table

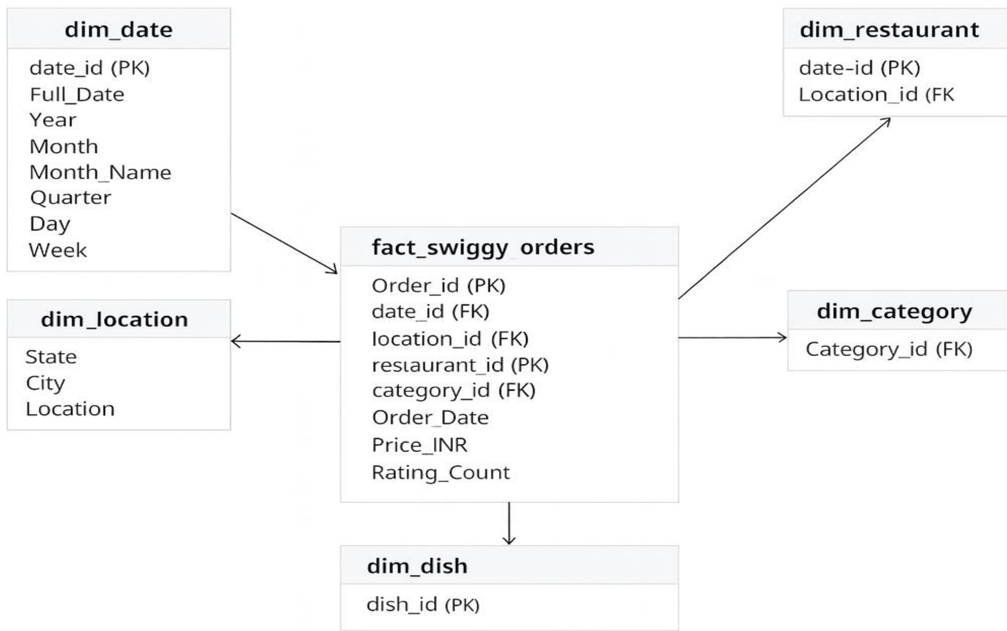
Central fact table: **`fact_swiggy_orders`**

- Measures: Price_INR, Rating, Rating_Count
- Foreign keys: Links to all dimension tables (date, location, restaurant, category, dish)

Populate each dimension with distinct data from the cleaned source and load the fact table with all foreign keys resolved.

2.3 Entity-Relationship Diagram (Star Schema)

The following diagram illustrates the star schema design with the central fact table surrounded by dimension tables:



Star Schema for Swothy dataset (fact_swchy_orders wth dim_date, dim_location, dim_category)

3. KPI Development

Once the star schema is built, compute the following core performance indicators to drive business insights.

3.1 Basic KPIs

KPI	Description
Total Orders	Count of all unique orders in the dataset
Total Revenue (INR Million)	Sum of Price_INR converted to millions
Average Dish Price	Mean price of dishes across all orders
Average Rating	Mean customer rating across dishes

4. Deep-Dive Business Analysis

4.1 Date-Based Analysis

- Monthly order trends
- Quarterly order trends
- Year-wise growth
- Day-of-week patterns
- Month that generates the highest revenue
- Month-over-month revenue growth

4.2 Location-Based Analysis

- Top 10 cities by order volume
- Revenue contribution by state
- Cities with the highest average-rated restaurants
- Percentage of total revenue contributed by each state
- Restaurants Swiggy should prioritize for promotion
- Identification of high-value restaurants

4.3 Food Performance

- Top 10 restaurants by order volume
- Top categories (Indian, Chinese, etc.)
- Most ordered dishes
- Cuisine performance: Orders + Average Rating
- Restaurants that contribute the most revenue
- Restaurants with high ratings but low sales (opportunity analysis)

4.4 Customer Spending Insights

Analyze order distribution across the following customer spend buckets:

Spend Bucket (INR)	Description
Under 100	Low-value orders
100 – 199	Budget-conscious orders
200 – 299	Mid-range orders
300 – 499	Premium orders
500+	High-value / large orders

Report total order distribution across these ranges to understand customer spending behavior.

4.5 Ratings Analysis

Analyze the distribution of dish ratings from 1 to 5. Identify patterns in customer satisfaction and correlate ratings with order volume and revenue where relevant.

End of Business Requirements Document